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Experiment No. 03

Aim : To deploy multi-service applications using Docker Compose

LO Mapped : LO4

LO : Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory :

What is Docker Compose ?

Docker Compose is a tool that allows you to define, configure, and manage multiple Docker containers using a single configuration file called docker-compose.yml. Instead of running each container separately with multiple docker run commands, Docker Compose enables you to start, stop, and manage all the services together using one command.

A Docker Compose file is written in YAML (YAML Ain't Markup Language) format and contains the complete configuration of the application, including container images, networking, ports, environment variables, dependencies, and storage volumes.

In this experiment, Docker Compose is used to deploy a multi-service application consisting of four containers:

- Nginx – Web Server
- MySQL – Database Server
- phpMyAdmin – Database Management Interface
- Redis – In-Memory Cache

All these services communicate with each other through Docker's internal network.

Why Use Docker Compose?

Docker Compose is a tool used to define and manage multi-container Docker applications using a single docker-compose.yml file. It simplifies deployment by allowing all services to be started, stopped, and managed with one command.

Benefits of Docker Compose

1. Centralized Configuration: All container settings such as images, ports, environment variables, and volumes are defined in one YAML file, making the application easy to configure and maintain.

2. Automatic Networking: Docker Compose automatically creates a private network where all services can communicate using their service names (e.g., phpmyadmin connects to MySQL using database).
3. Persistent Storage: Named volumes store important data outside the container, ensuring that database data is not lost when containers are restarted or recreated.
4. Easy Deployment: Multiple services can be started, stopped, or removed using simple commands like docker compose up -d and docker compose down.

Creating the docker-compose.yml File

Step 1: Create a Project Directory

A separate project folder is created to store the Docker Compose configuration.

```
mkdir docker-compose-lab
```

Move into the project directory.

```
cd docker-compose-lab
```

Step 2: Create the YAML File

Create the Docker Compose configuration file.

```
notepad docker-compose.yml
```

A new file named **docker-compose.yml** opens in Notepad.

Docker Compose File Structure

The experiment uses the following Docker Compose configuration:

```
version: "3.8"
services:
  web:
    image: nginx:latest
    container_name: webserver
    ports:
      - "8082:80"
    depends_on:
      - database

  database:
```

```
    image: mysql:8.0
    container_name: mysqlserver
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
      MYSQL_PASSWORD: student123
    ports:
      - "3307:3306"
    volumes:
      - mysql-data:/var/lib/mysql

  phpmyadmin:
    image: phpmyadmin/phpmyadmin
    container_name: phpmyadmin
    depends_on:
      - database
    environment:
      PMA_HOST: database
      MYSQL_ROOT_PASSWORD: root123
    ports:
      - "8083:80"

  redis:
    image: redis:latest
    container_name: redisserver
    ports:
      - "6380:6379"

volumes:
  mysql-data
```

Commands Used and Their Description

Step	Command	Description
1	docker --version	Checks whether Docker is installed and displays the installed Docker version.
2	docker compose version	Verifies that Docker Compose is installed and displays its version.
3	mkdir docker-compose-lab	Creates a new project directory named **docker-compose-lab** to store the Docker Compose files.
4	cd docker-compose-lab	Changes the current working directory to the **docker-compose-lab** folder.
5	notepad docker-compose.yml	Opens Notepad to create or edit the **docker-compose.yml** configuration file.
6	dir	Lists all files and folders in the current directory to confirm that the **docker-compose.yml** file has been created.
7	docker compose config	Validates the syntax and configuration of the **docker-compose.yml** file and displays the processed configuration.
8	docker compose up -d	Pulls the required images (if not available), creates the containers, and starts all services in detached (background) mode.
9	docker compose ps	Displays the status of all containers managed by Docker Compose, including their names, states, and port mappings.
10	docker compose images	Lists all Docker images being used by the services in the Docker Compose project.
11	docker compose logs	Displays the logs generated by all running containers for monitoring and troubleshooting.
12	docker compose logs -f	Continuously streams container logs in real time until the command is stopped.
13	docker ps	Lists all currently running Docker containers along with their IDs, images, status, and ports.
14	docker compose stop	Stops all running containers without removing them from the system.
15	docker compose start	Starts previously stopped containers without recreating them.
16	docker compose restart	Restarts all containers by stopping and then starting them again.
17	docker compose down	Stops and removes all containers and networks created by the Docker Compose project while preserving named volumes.
18	docker compose down -v	Stops and removes all containers, networks, and named volumes, deleting persistent data stored in the volumes.

Execution :

```
C:\Users\15L>docker --version
Docker version 29.6.1, build 8900f1d

C:\Users\15L>docker compose version
Docker Compose version v5.3.0
```

```
C:\Users\15L>mkdir docker-compose-lab

C:\Users\15L>cd docker-compose-lab

C:\Users\15L\docker-compose-lab>notepad docker-compose.yml

C:\Users\15L\docker-compose-lab>dir
 Volume in drive C is Windows
 Volume Serial Number is C84C-AF14

 Directory of C:\Users\15L\docker-compose-lab

28-07-2026  10:27    <DIR>          .
28-07-2026  10:27    <DIR>          ..
28-07-2026  10:27               816 docker-compose.yml
               1 File(s)            816 bytes
               2 Dir(s)  757,969,608,704 bytes free
```

```
C:\Users\15L\docker-compose-lab>docker compose config
time="2026-07-28T10:28:32+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
name: docker-compose-lab
services:
  database:
    container_name: mysqlserver
    environment:
      MYSQL_DATABASE: studentdb
      MYSQL_PASSWORD: student123
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_USER: student
    image: mysql:8.0
    networks:
      default: null
    ports:
      - mode: ingress
        target: 3306
        published: "3306"
        protocol: tcp
    volumes:
      - type: volume
        source: mysql-data
        target: /var/lib/mysql
        volume: {}
  phpmyadmin:
    container_name: phpmyadmin
    depends_on:
      database:
        condition: service_started
        required: true
    environment:
      MYSQL_ROOT_PASSWORD: root123
      PMA_HOST: database
    image: phpmyadmin/phpmyadmin
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8081"
        protocol: tcp
  redis:
    container_name: redisserver
    image: redis:latest
    networks:
      default: null
```

```
C:\Users\15L\docker-compose-lab>docker compose up -d
time="2026-07-28T10:44:12+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
[+] up 4/4
 ✔Container mysqlserver Running                                                                  0.0s
 ✔Container redisserver Started                                                                  1.5s
 ✔Container phpmyadmin  Started                                                                  1.5s
 ✔Container webserver   Started                                                                  1.5s
```

```
C:\Users\15L\docker-compose-lab>docker compose ps
time="2026-07-28T11:43:38+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
NAME          IMAGE                   COMMAND                  SERVICE      CREATED          STATUS
      PORTS
mysqlserver   mysql:8.0               "docker-entrypoint.s…"   database     53 minutes ago   Up 53 min
utes   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin    phpmyadmin/phpmyadmin   "/docker-entrypoint.…"   phpmyadmin   53 minutes ago   Up 53 min
utes   0.0.0.0:8083->80/tcp, [::]:8083->80/tcp
redisserver   redis:latest            "docker-entrypoint.s…"   redis        53 minutes ago   Up 53 min
utes   0.0.0.0:6380->6379/tcp, [::]:6380->6379/tcp
webserver     nginx:latest            "/docker-entrypoint.…"   web          53 minutes ago   Up 53 min
utes   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
```

```
C:\Users\15L\docker-compose-lab>docker compose images
time="2026-07-28T10:52:29+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
CONTAINER          REPOSITORY             TAG                 PLATFORM           IMAGE ID
  SIZE               CREATED
mysqlserver        mysql                  8.0                 linux/amd64        6cd09145362d
  799MB              2 months ago
phpmyadmin         phpmyadmin/phpmyadmin  latest              linux/amd64        e66b1f5a8c58
  742MB              9 months ago
redisserver        redis                  latest              linux/amd64        3848bb827e71
  140MB              4 months ago
webserver          nginx                  latest              linux/amd64        4e5db4761e0f
  161MB              12 days ago
```

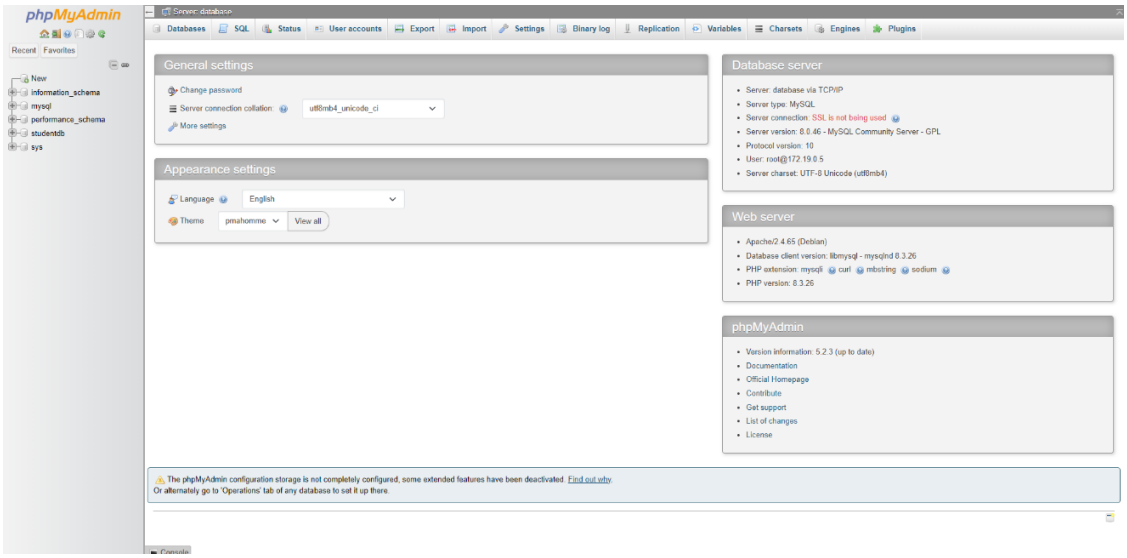
```
C:\Users\15L\docker-compose-lab>docker compose restart
time="2026-07-28T10:48:41+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
[+] restart 0/4
 - Container mysqlserver Restarting                                                              1.8s
 - Container webserver   Restarting                                                              1.8s
 - Container phpmyadmin  Restarting                                                              1.8s
 - Container redisserver Restarting                                                              1.8s
```

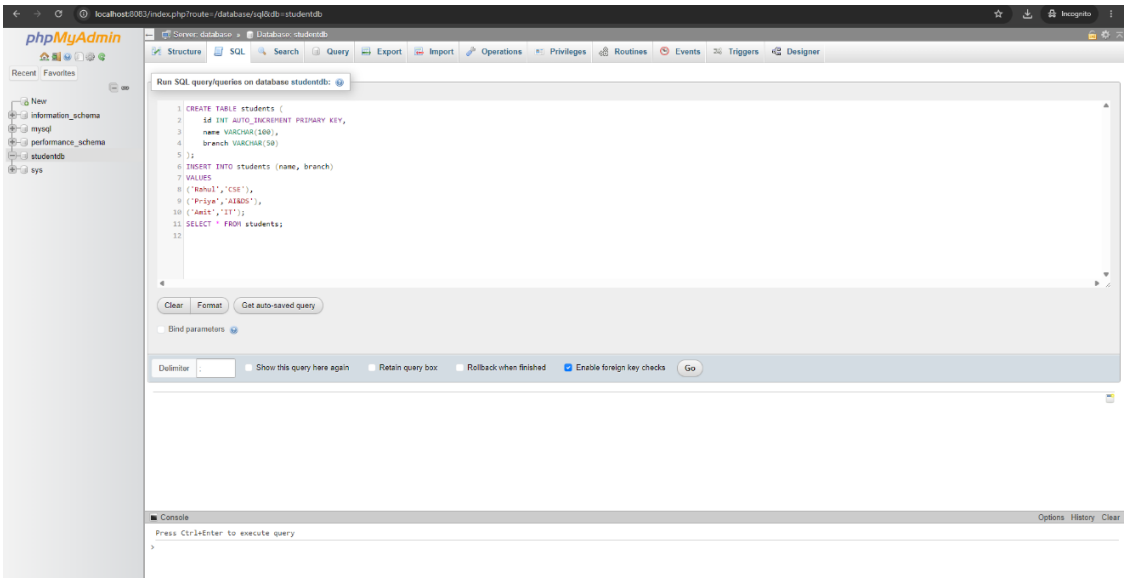
```
C:\Users\15L\docker-compose-lab>docker compose down
time="2026-07-28T10:48:56+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
[+] down 5/5
 ✔Container webserver               Removed                                                      0.4s
 ✔Container phpmyadmin              Removed                                                      1.3s
 ✔Container redisserver             Removed                                                      0.5s
 ✔Container mysqlserver             Removed                                                      1.1s
 ✔Network docker-compose-lab_default Removed                                                     0.3s
```

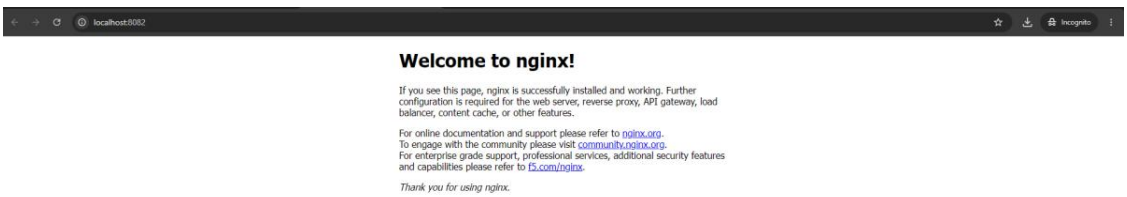
```
C:\Users\15L\docker-compose-lab>docker compose down -v
time="2026-07-28T10:49:11+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
[+] down 1/1
 ✔Volume docker-compose-lab_mysql-data Removed                                                  0.0s

C:\Users\15L\docker-compose-lab>docker compose up
time="2026-07-28T10:49:48+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
[+] up 6/6
 ✔Network docker-compose-lab_default   Created                                                  0.0s
 ✔Volume docker-compose-lab_mysql-data Created                                                  0.0s
 ✔Container mysqlserver                Created                                                  0.0s
 ✔Container redisserver                Created                                                  0.0s
 ✔Container webserver                  Created                                                  0.0s
 ✔Container phpmyadmin                 Created                                                  0.0s
Attaching to mysqlserver, phpmyadmin, redisserver, webserver
mysqlserver  | 2026-07-28 05:19:48+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.4
6-1.el9 started.
redisserver  | Starting Redis Server
redisserver  | 1:C 28 Jul 2026 05:19:48.522 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
redisserver  | 1:C 28 Jul 2026 05:19:48.522 * Redis version=8.6.2, bits=64, commit=00000000, modified=
1, pid=1, just started
redisserver  | 1:C 28 Jul 2026 05:19:48.522 * Configuration loaded
redisserver  | 1:M 28 Jul 2026 05:19:48.522 * monotonic clock: POSIX clock_gettime
redisserver  | 1:M 28 Jul 2026 05:19:48.523 * Running mode=standalone, port=6379.
redisserver  | 1:M 28 Jul 2026 05:19:48.524 * <bf> RedisBloom version 8.6.0 (Git=unknown)
redisserver  | 1:M 28 Jul 2026 05:19:48.524 * <bf> Registering configuration options: [
redisserver  | 1:M 28 Jul 2026 05:19:48.524 * <bf>      { bf-error-rate       :      0.01 }
redisserver  | 1:M 28 Jul 2026 05:19:48.524 * <bf>      { bf-initial-size     :       100 }
redisserver  | 1:M 28 Jul 2026 05:19:48.524 * <bf>      { bf-expansion-factor :         2 }
```

```
C:\Users\15L\docker-compose-lab>docker compose logs
time="2026-07-28T11:44:26+05:30" level=warning msg="C:\\Users\\15L\\docker-compose-lab\\docker-compose
.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential con
fusion"
phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
 using 172.19.0.5. Set the 'ServerName' directive globally to suppress this message
phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
 using 172.19.0.5. Set the 'ServerName' directive globally to suppress this message
phpmyadmin  | [Tue Jul 28 05:19:48.688608 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4
.65 (Debian) PHP/8.3.26 configured -- resuming normal operations
phpmyadmin  | [Tue Jul 28 05:19:48.688661 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'ap
ache2 -D FOREGROUND'
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET / HTTP/1.1" 200 6197 "-" "Mozilla/5.0 (
Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /themes/pmahomme/jquery/jquery-ui.css H
TTP/1.1" 200 8823 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.
36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/vendor/codemirror/lib/codemirror.cs
s?v=5.2.3 HTTP/1.1" 200 2847 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Apple
WebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/vendor/codemirror/addon/hint/show-h
int.css?v=5.2.3 HTTP/1.1" 200 668 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/vendor/codemirror/addon/lint/lint.c
ss?v=5.2.3 HTTP/1.1" 200 1634 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) Appl
eWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/dist/keyhandler.js?v=5.2.3 HTTP/1.1
" 200 1116 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHT
ML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/vendor/jquery/jquery-migrate.min.js
?v=5.2.3 HTTP/1.1" 200 5401 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleW
ebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/dist/ajax.js?v=5.2.3 HTTP/1.1" 200
8188 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, li
ke Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/dist/name-conflict-fixes.js?v=5.2.3
 HTTP/1.1" 200 331 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537
.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/vendor/js.cookie.js?v=5.2.3 HTTP/1.
1" 200 1869 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KH
TML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.19.0.1 - - [28/Jul/2026:05:25:11 +0000] "GET /js/vendor/jquery/jquery.min.js?v=5.2.3
 HTTP/1.1" 200 30709 "http://localhost:8083/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/5
37.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
```


phpMyAdmin
Server: database
Databases
SQL
Status
User accounts
Export
Import
Settings
Binary log
Replication
Variables
Charsets
Engines
Plugins
General settings
Change password
Server connection collation:
utf8mb4_unicode_ci
More settings
Appearance settings
Language
English
Theme
pmahomme
View all
Database server
Server: database via TCP/IP
Server type: MySQL
Server connection: SSL is not being used
Server version: 8.0.46 - MySQL Community Server - GPL
Protocol version: 10
User: root@172.19.0.5
Server charset: UTF-8 Unicode (utf8mb4)
Web server
Apache/2.4.65 (Debian)
Database client version: libmysql - mysqlnd 8.3.26
PHP extension: mysqli curl mbstring sodium
PHP version: 8.3.26
phpMyAdmin
Version information: 5.2.3 (up to date)
Documentation
Official Homepage
Contribute
Get support
List of changes
License
The phpMyAdmin configuration storage is not completely configured, some extended features have been deactivated. Find out why.
Or alternately go to 'Operations' tab of any database to set it up there.
Recent
Favorites
New
information_schema
mysql
performance_schema
studentdb
sys
Console


localhost:8083/index.php?route=/database/sql&db=studentdb
phpMyAdmin
Server: database » Database: studentdb
Structure
SQL
Search
Query
Export
Import
Operations
Privileges
Routines
Events
Triggers
Designer
Run SQL query/queries on database studentdb:
CREATE TABLE students (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100),
branch VARCHAR(50)
);
INSERT INTO students (name, branch)
VALUES
('Rahul','CSE'),
('Priya','AI&DS'),
('Amit','IT');
SELECT * FROM students;
Clear
Format
Get auto-saved query
Bind parameters
Delimiter
Show this query here again
Retain query box
Rollback when finished
Enable foreign key checks
Go
Console
Options
History
Clear
Press Ctrl+Enter to execute query
Recent
Favorites
New
information_schema
mysql
performance_schema
studentdb
sys


localhost:8082
Welcome to nginx!
If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.
For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.
Thank you for using nginx.

Conclusion :

In this experiment, a multi-tier application stack comprising MySQL, Adminer, Nginx, and Redis was successfully orchestrated using Docker Compose. By defining services, port mappings, environment variables, and persistent storage in a single docker-compose.yml manifest, the process of configuring and managing complex multi-container environments was streamlined compared to running individual container commands. Interactive terminal verification confirmed service health across all mapped endpoints.